

Dongwei(James) Zhang

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

M.Eng. in Financial Engineering with Data Science Certificate

Expected Dec 2025

- **GPA: 3.6; Coursework:** Operation Research, Market Microstructure, Derivatives, Portfolio Management

University of Southern California

Los Angeles, CA

B.S. in Applied and Computational Mathematics, Minor in Computer Science

May 2024

- **GPA: 3.7; Coursework:** Dynamic Modeling, Optimization, Deep Learning, Algorithms, Data Structures

SKILLS

Technical: Python(NumPy, Pandas, scikit-learn, Keras), C/C++, R, Machine Learning, MATLAB, SQL, Java

EXPERIENCE

AbleMarkets

New York, NY

Quantitative Research Analyst

Nov 2024 – Present

- Co-authored "*Pricing Bitcoin and Other Assets with Conditional Optimization*" (SSRN), introducing a nonlinear pricing model achieving **100% excess returns** over spot price through volume-based predictions
- Extended network adoption models to multi-asset portfolio optimization, resulting in an **11.56%** annualized return and a Sharpe ratio of **1.196**, outperforming traditional models (0.35% return, 0.232 Sharpe ratio)

CITIC Securities

Beijing, China

Quantitative Research Summer Analyst

May – July 2024

- Researched a multi-factor stock selection strategy and its performance in the A-share market; applied regression analysis to identify factors for returns; achieved an increase in annual returns from **12%** to **18.44%**
- Developed an automated data pipeline to pull financial data from Wind database; performed premium analysis between China A-share and Hong Kong H-share of cross-listed companies
- Developed a mid-frequency regression statistical arbitrage strategy with a backtested Sharpe ratio of **2.09** for A-H double-listed equities from 2022-2024 in 8 industries

Northern Purchase

Los Angeles, CA

Quantitative Analyst

Sep – Dec 2023

- Applied Monte Carlo simulations; built revenue models with 12 variables to stress-test earnings volatility under 10+ varied economic conditions, yielding a comprehensive risk-adjusted valuation framework
- Developed VBA macros to streamline data extraction and preprocessing from financial statements, improving processing efficiency by **35%** and enabling dynamic risk scenario modeling

Apex Capital Growth Advisors

Los Angeles, CA

Quantitative Analyst

July – Sep 2023

- Applied Principal Component Analysis(PCA) to identify fundamental key factors; leveraged PCA model to rank companies based on adjusted cash flow performance
- Conducted industry research to assess growth prospects and competitive landscape, evaluating each company against criteria such as business model scalability, management strength, and recurring revenue consistency

PROJECTS

SPX Options Trading Strategy with Delta Hedging

Ithaca, NY

Cornell Trading Competition https://github.com/DongweiZhang/trading_machine_European-style_SPX

Aug – Nov 2024

- Developed a delta-neutral low-frequency trading strategy utilizing complex option structures (Straps, Strips, Condors, and Strangles) to exploit volatility signals
- Achieved a 3-month return of **17%** with a Sharpe ratio of **2.54** and a maximum drawdown of **0.016** on minute-level data from April to June 2024; ranked **Top 10%** in the competition

ACTIVITIES & INTERESTS

President of USC Board Game Club, Poker, Strategic Games, Puzzle, Soccer, Cello, Film Noir, Traveling